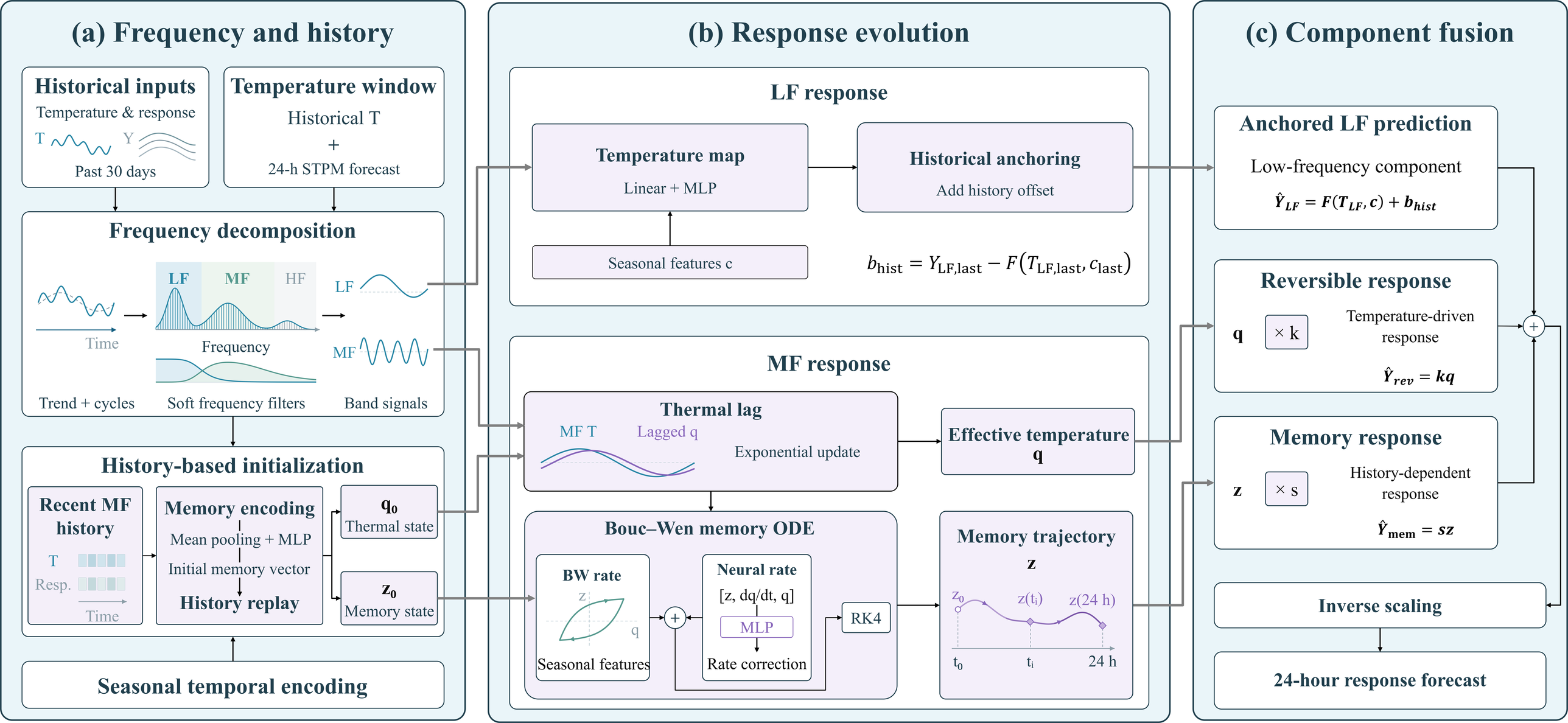

(a) Frequency and history
Historical inputs
Temperature & response
T
Y
Past 30 days
Temperature window
Historical T
+
24-h STPM forecast
Frequency decomposition
LF
MF
HF
Time
Frequency
LF
MF
Trend + cycles
Soft frequency filters
Band signals
History-based initialization
Recent MF history
T
Resp.
Time
Memory encoding
Mean pooling + MLP
Initial memory vector
History replay
q0
Thermal state
z0
Memory state
Seasonal temporal encoding
(b) Response evolution
LF response
Temperature map
Linear + MLP
Historical anchoring
Add history offset
Seasonal features c
b_hist = Y_LF,last − F(T_LF,last, c_last)
MF response
Thermal lag
MF T
Lagged q
Exponential update
Effective temperature q
Bouc–Wen memory ODE
BW rate
z
q
Seasonal features
Neural rate
[z, dq/dt, q]
MLP
Rate correction
RK4
Memory trajectory
z
z0
z(ti)
z(24 h)
t0
ti
24 h
(c) Component fusion
Anchored LF prediction
Low-frequency component
Ŷ_LF = F(T_LF, c) + b_hist
Reversible response
q
× k
Temperature-driven response
Ŷ_rev = kq
Memory response
z
× s
History-dependent response
Ŷ_mem = sz
Inverse scaling
24-hour response forecast